GSSS INSTITUTE OF ENGINEERING & TECHNOLOGY FOR WOMEN, MYSURU
Department of Electronics & Communication Engineering | GATE Preparation Library

General Aptitude — Official GATE ECE Syllabus

Paper Code: EC • Section: General Aptitude (Mandatory 15 Marks for All Candidates)

Weightage: 15 Marks Fixed • **Total Questions:** 10 Questions (5x1M + 5x2M) • **Score Impact:** Decisive for Top Rank

OFFICIAL GATE EXAMINATION PATTERN & MARKING SCHEME
• **Question Types:** Multiple Choice Questions (MCQ, 1/3 negative marks), Multiple Select Questions (MSQ, zero negative marks), Numerical Answer Type (NAT, zero negative marks).
• **Weightage:** General Aptitude (15 Marks) + Engineering Mathematics (13 Marks) + Core Technical Subjects (72 Marks) = **100 Total Marks (65 Questions, 180 Minutes)**.

SECTION: VERBAL APTITUDE

- Basic English Grammar: Tenses, subject-verb agreement, noun-pronoun agreement, active and passive voice, direct and indirect speech.
- Parts of speech: Articles, prepositions, conjunctions, conditional clauses, modal verbs.
- Vocabulary in context: Synonyms, antonyms, homonyms, phrasal verbs, frequently confused words.
- Idioms and Phrases: Meanings, contextual usage, sentence completion.
- Reading comprehension: Factual and inferential comprehension passages, theme detection, identifying tone and purpose of author.
- Narrative sequencing: Sentence rearrangement (para-jumbles), logical flow and coherence of paragraphs.

SECTION: QUANTITATIVE APTITUDE

- Data Interpretation: Data tables, bar charts, pie charts, line graphs, two- and three-dimensional plots, Venn diagrams.
- Numerical computation and estimation: Ratios, proportions, percentages, profit and loss, simple and compound interest.
- Powers, exponents, logarithms, surds, indices, factorization, linear and quadratic equations.
- Permutations and Combinations: Fundamental counting principle, arrangements, selections, circular permutations.
- Probability: Classical probability, addition and multiplication rules, conditional probability, independent events, Bayes' theorem applications.
- Sequences and series: Arithmetic Progression (AP), Geometric Progression (GP), Harmonic Progression (HP), sum of series.
- Speed, Time, and Distance: Average speed, relative speed, trains, boats and streams, circular tracks.
- Work and Time: Man-days concept, individual and collaborative work, pipes and cisterns.
- Geometry and Mensuration: Lines, angles, triangles, polygons, circles. Surface areas and volumes of 2D and 3D shapes (cylinders, spheres, cones, prisms).

SECTION: ANALYTICAL APTITUDE

- Logic and deduction: Deductive and inductive reasoning, conditional logic (if-then statements, contrapositive, converse, inverse).
- Syllogisms: Statements and conclusions, validity of categorical syllogisms using Venn diagrams.
- Analogy and relationship identification: Word analogies, number analogies, symbol relationships.
- Blood relations, family trees, direction sense test.
- Seating arrangements: Linear arrangements, circular arrangements, matrix/tabular grid puzzles.
- Coding and decoding: Letter coding, number coding, substitution coding.
- Critical reasoning: Weakening and strengthening arguments, assumptions, inferences, course of action.

SECTION: SPATIAL APTITUDE

- Transformation of 2D and 3D shapes: Translation, rotation (clockwise and counter-clockwise), scaling, reflection, mirroring.
- Assembling and grouping shapes: Matching identical geometric shapes, combining 2D patterns into 3D objects.
- Paper folding, punching, and cutting: Predicting unfolded paper patterns.
- Cubical blocks and dice: Opposite faces, patterns on folded cubes, counting hidden faces/blocks.
- Pattern identification: Identifying repeating patterns, non-verbal series continuation, figure matrix completions.

RECOMMENDED STANDARD TEXTBOOKS & REFERENCES

- A Modern Approach to Verbal & Non-Verbal Reasoning — R.S. Aggarwal
- Quantitative Aptitude for Competitive Examinations — R.S. Aggarwal
- Word Power Made Easy — Norman Lewis

HIGH-YIELD GATE EXAM STRATEGY & REVISION TIPS

- General Aptitude contributes 15% of the total score and has the highest marks-to-time ratio in GATE.
- All 10 questions can be comfortably solved in 15-18 minutes during the actual exam.
- Spatial aptitude questions have been mandatory since GATE 2021; always practice 3D folded cube transformations.